FIRST COURSE HANDOUT,
PROBABILITY THEORY (MTH309A),
2023-24 EVEN SEMESTER
3 - 1 - 0 - 0 [11 CREDITS]

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1. Pre-requisites and Course Objectives

This is a compulsory course for the first year Masters' (Statistics) students. Familiarity with Real Analysis will be assumed. A prior course on basic Probability distributions is a prerequisite (MSO201/MSO205/MTH431 or equivalent).

Objectives: First half of the course shall focus on Brownian motion and Itô Stochastic integration. In the later half, we shall discuss Stochastic Differential Equations and its applications. At the end of the course, the students are expected to have enough familiarity with the subject to apply the results in their own fields of study.

2. Course contents and References

2.1. **Course contents.**

(Topic 1) σ-field of events (emphasis on σ-fields generated by a collection of sets, Borel σ-fields), Limits of sequences of sets. Principle of good sets (Monotone Class Theorem, Dynkin's $\pi - \lambda$ Theorem).

(Topic 2) Measure and Measure space (emphasis on probability measure and probability space), properties, Borel-Cantelli Lemma (first half).

(Topic 3) Extension of measures (existence and uniqueness), Outer Measure, Carathéodery Extension Theorem for σ-finite measure. Completeness of measure spaces.

(Topic 4) Measurable functions (emphasis on Random variables), algebraic properties and limits of measurable functions. Probability distribution of Random variables, induced probability space, distribution function.

(Topic 5) Correspondence between distribution functions and Lebesgue-Stieltjes measures (emphasis on probability measures), construction of Lebesgue measure.

(Topic 6) Integration of measurable functions (emphasis on Expectation and moments of Random variables, change of variables/measures), inequalities.

(Topic 7) Convergence theorems for expectations of sequences of random variables (monotone convergence theorem, Fatou's lemma, dominated convergence theorem).

(Topic 8) Connection between Riemann and Lebesgue Integration.

(Topic 9) Various modes of convergence of sequences of random variables (almost surely, in probability, in r-th mean). L^p spaces.

(Topic 10) Independence of events and random variables. Borel-Cantelli Lemma (second half).

Date: January 3, 2023.

(Topic 11) Convergence of series of independent random variables, Kolmogorov inequality, Kolmogorov three-series criterion.

(Topic 12) Characteristic function and its properties, inversion formulae.

(Topic 13) Weak convergence of probability measures, connection with convergence of sequences of distribution functions, Helly-Bray theorems, Portmanteu Theorem.

(Topic 14) Khintchin's weak law of large numbers, Kolmogorov strong law of large numbers. Central limit theorems of Lindeberg-Levy, Liapounov and Lindeberg-Feller.

2.2. **Reference materials.** References:

- Probability & Measure Theory (2nd Edition), Robert B. Ash and Catherine A. Doléans-Dade. Elsevier.
- A Course in Probability Theory (3rd Edition), Kai Lai Chung. Academic Press (Elsevier).
- Probability and Measure (3rd Edition), Patrick Billingsley. Wiley.

Supplimentary texts:

- Probability: Theory and Examples (4th Edition), Rick Durrett. CUP.
- Probability Essentials (2nd Edition), Jean Jacod and Philip Protter. Springer.
- Probability with Martingales, David Williams, CUP.
- Modern Probability Theory (3rd Edition), B. R. Bhat, New Age International.
- Probability Theory - I (4th Edition), M Loéve, Springer.

2.3. **Important dates.**

Week #	Dates	Remarks
1	January 4 - 5	Start of the course
2	January 8 - 12	
3	January 15 - 19	
4	January 22 - 26	Holiday: January 26
5	January 29 - February 2	
6	February 5 - 9	
7	February 12 - 16	
Mid Sem Exam	February 19 - 24	As per schedule announced by DOAA
8	February 26 - March 1	
9	March 4 - 8	Holiday: March 8
10	March 11 - 15	
11	March 18 - 22	Drop deadline: March 22
Mid Sem Recess	March 23 - 31	
12	April 1 - 5	
13	April 8 - 12	Holiday: April 11
14	April 15 - 19	End of the course
End Sem Exam	April 22 - May 1	As per schedule announced by DOAA
Make-up Exam	May 4 - 6	As per schedule announced by DOAA

3. Lecture Schedule and Office Hours

- Lectures: Tuesdays, Thursdays and Fridays 8:00 - 8:50 hrs (supplementary materials may be made available over helloIITK portal).
- Tutorial: Wednesdays 8:00 - 8:50 hrs
- Lecture and Tutorial venue: L04
- Office Hours: Students should communicate their need to discuss doubts by email and take appointments for the same.

4. Weightages for different components of evaluation (out of 100)

Component Name	Weightage
Mid-semester Examination	28
End-semester Examination	42
Quiz (Best of 2 out of 3)	2x15 = 30

- Mid/End Semester Examination: The examinations shall be conducted as per the schedule announced by DOAA. The syllabus, question pattern and other details shall be announced later on.
- Quiz: There shall be 3 Quizzes (15 marks in each) and the best 2 of these 3 shall be used in grading. Syllabus and other details of the quizzes shall be announced from time to time. Neither a make-up opportunity nor proration shall be provided for a missed quiz.
- Exercises/Practice Problems: Exercises and/or practice problems shall be provided on a weekly basis and students are expected to solve them on their own. There is no need to submit the solutions. However, students are strongly encouraged to discuss amongst themselves and also solve problems from the books mentioned above.

5. Course Policies: Attendance, Grading, Withdrawal

- There is no extra weightage for attendance. However, students are strongly advised to attend all the lectures.
- The policy of relative grading will be followed for awarding final grades. However, if the highest marks is below 90, then A* grade may not be awarded.
- No make-up opportunity shall be provided for a missed Mid Semester Examination or a quiz.
- In the case where a student misses the Mid-Semester examination due to a bonefide non-emergency situation, then a request for proration may be considered. However, such a request must be made well in advance and the final decision to allow prorating rests with the instructor.
- For missing the Mid Semester examination due to an emergent situation, request for proration must be made as soon as possible after the exam date. In case of a medical emergency, the student must present a letter from the doctor stating that the student was not in a condition to take the examination. The final decision to allow prorating rests with the instructor.
- Make-up option for the End-semester examination will be as approved by DOAA. Applications for the same need to be supported with proof, such as, a medical certificate or a proof of sanctioned leave.
- Any dishonest practice during examinations or quizzes will be reported to DOAA and appropriate action would be taken to penalize such action.
- Students are allowed to withdraw from the course as per the guidelines set by DOAA.